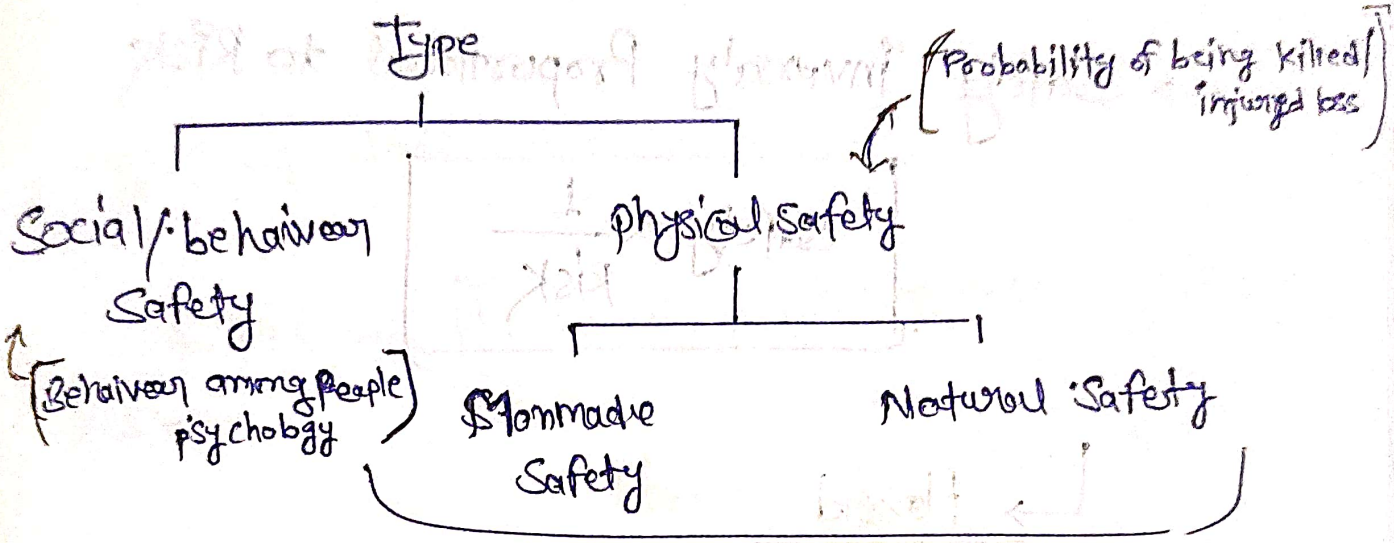
Safety

★ Something that is free from Risk, harm or injury.



⇒ It is state of being safe from risk, harm hazard or injury. it is used to denote risk.

(i) inventify the risk
(ii) potential cause
(iii) Eluminate those cause.

Risk

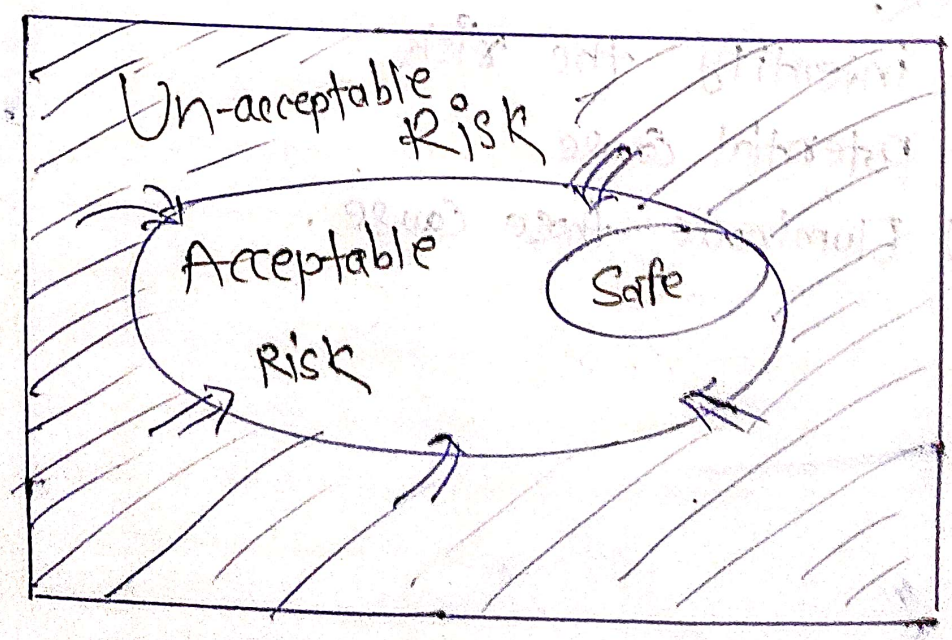
* Risk may be defined as the hazard associated with an event, process or system.

→ Safety Inversely Proportional to Risk

$$\text{Safety} \propto \frac{1}{\text{Risk}}$$

↳ Hazard

Where, Hazard is the potential intent to cause harm to the human body in the form of illness, injury or totality or to the system changing its state.



⇒ Double definition of Risk —

$$\underset{\text{(Risk Rating)}}{\text{Risk}} = \underbrace{\left[\begin{array}{c}\text{Probability of} \\ \text{occurrence of an} \\ \text{Event}\end{array}\right] \times \left[\text{Consequence}\right]}_{\text{Two Component}}$$

⇒ Triplet definition of Risk —

$$\underset{\text{(Risk Value)}}{\text{Risk}} = \underbrace{\left[\text{Probability}\right] \times \left[\text{Consequence}\right] \times \left[\text{Exposure}\right]}_{\text{Three Component}}$$

⇒ **Risk ↑ Safety ↓ Accident ↑**

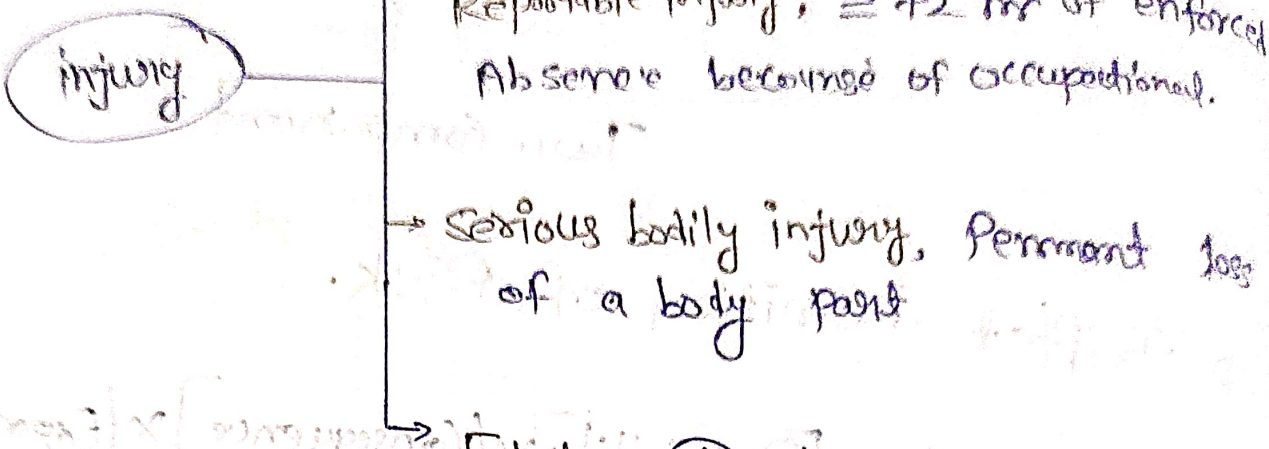
★ **Accident and Incident ⇒**

Accident – An unplanned, unwanted and unexpected event, occuring through a sequence of events, leading to an undesirable.

Incindent – An unplanned, unwanted & unexpected event, occuring through a sequence of events, may or may not leading to an undesirable consequences.

→ Danger occurance.

☆ Injury ⇒ (24 ≤ x < 72)



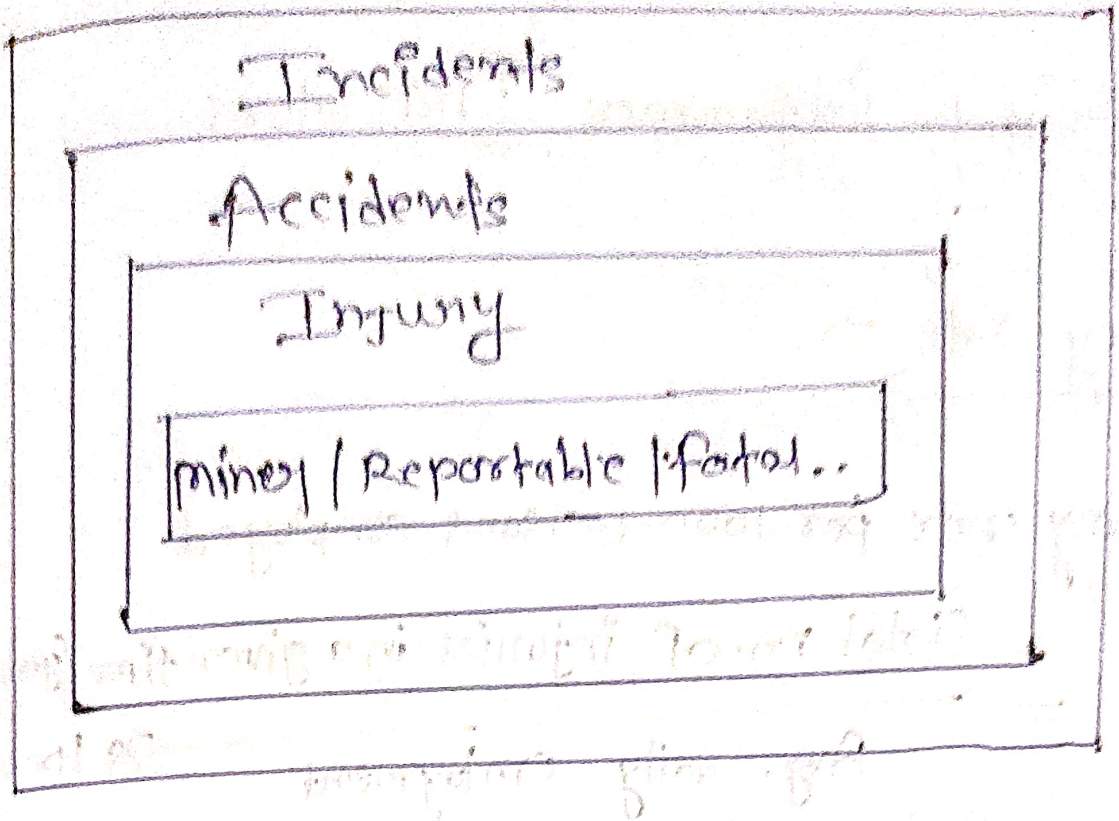
Disaster - Dis - aster = ~~Disaster~~ Disaster

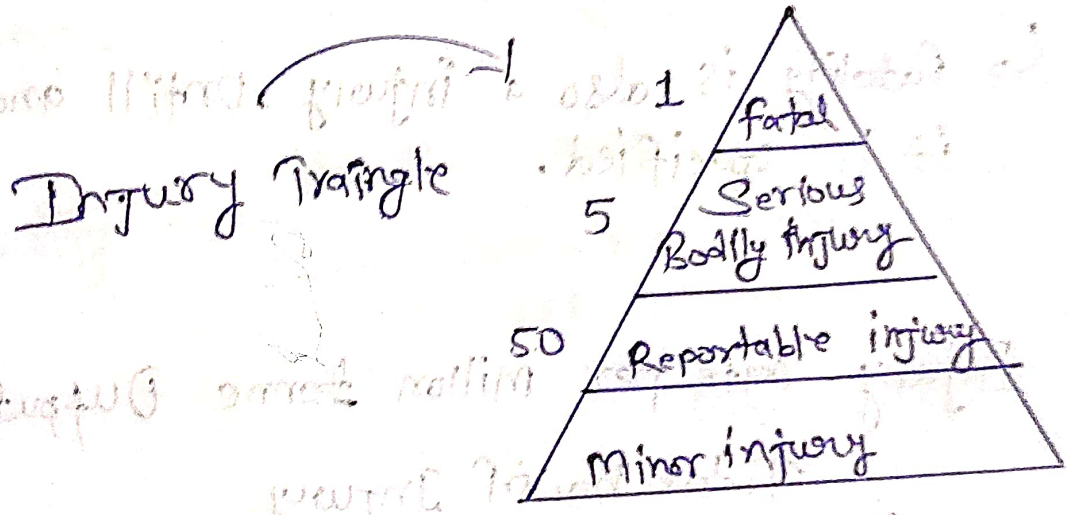
(Bad) (Star)

↳ more than 10 fatal injuries in an accident thus we say disaster.

↳ in a particular accident 10 or more person are died then it is known as disaster.

⇒ Major accident (fg > death > 4)


Incidents
Accidents
Injury
miner / Reportable / fatal..


Injury Traingle
1
fatal
5
Serious Bodily Injury
50
Reportable injury
Minor injury

Safety Performance Indicators

Frequency rate ⇒

① Injury rate per 1000 persons employed

$$= \frac{\text{Total no. of injuries in a given time (year)}}{\text{Avg. daily embyment}} \times 1000$$

↳ fatality is also a injury untill and unless it is specified.

② Injury rate per million tonne Output

$$= \frac{\text{Total no. of Injury}}{\text{Total production (te)}} \times 10^6$$

③ Injury rate per 3 lakh man shifts worked

$$= \frac{\text{Total no. of injuries in a given time period}}{\text{Total man shifts worked during that period}} \times 10^5$$

④ Injury rate per million man hours worked.

$$= \frac{\text{Total no. of injuries in a given time period}}{\text{Total man-hours worked}} \times 10^6$$

⑤ Severity Index (DGMS)

$$= \frac{\text{Sof} \neq S}{\text{Total man-shift worked}} \times 10^5$$

⑥ ASTC – American Standard Time charges

$$\text{Severity} = \frac{\text{Total time charges (as per ASTC)}}{\text{Total man-hours worked in that period}} \times 10^6$$

(like – 8 hr x day)

⑦ Accident Severity Rate (ASR)

$$= \frac{\text{Total time charges (in day)}}{\text{Total man-hours worked}} \times 10^6$$

① Frequency rate are reactive, not predictive

② Accidents are rare events and so frequency values on them may not be statistically valid.

③ Frequency rates can be manipulated

④ Too much forces on severity rates / Indices false.

Security.

☆ How Can you prepare a representative index. Prepare a model which will provide a index.

Domino Theory / Heinrich Philosophy

1931: Heirich → Engg. and inspection divisin of traveller's insurance Company publishid a book industrial accident prevention.

☆ Heinrich's philosophy for Causation of Accidents

Domino theory:—

→ Prior to 1931, it was believed that accidents are caused by unsafe condition.

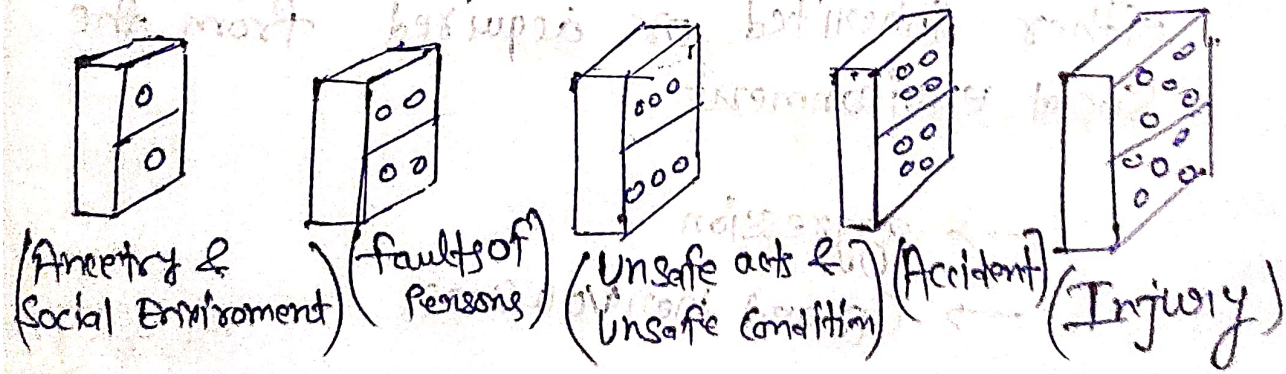
→ But in 1931 H.W. Heinrich published a book an accidents are mostly being caused by the unsafe acts of the people rather than the unsafe conditions.

→ After the people have emphasized on ill-human behaviour at workplace & to control the industrial accidents

☆ Domino :—

- A small rectangular block used in playing the game of dominoes, generally played with flat oblong pieces of wood.
- The face of each block has two equal areas that contains dots representing numbers.
- The dominoes are placed in such a manner that the fall of first domino causes the fall of the entire row, and if any one of the domonoes is removed, the series will interrupted

⇒ five factors in the injury sequence



⇒ 88 % — Unsafe acts
10% — Unsafe Condition
2% — Condition not classified (Act of god)

Domino Theory Explanation

⇒ ~~Ancestry and Social Environment~~

- Industrial injuries result only from accidents.
- Accidents are caused directly by the unsafe acts of persons and / or exposure to unsafe condition.
- Unsafe acts and conditions are caused only by faults of persons.
- faults of person are created by his social environment or acquired by inheritance.

⇒ Ancestry and Social Environment

- Undesirable of character may be either inherited or acquired from the social environment.
  - → aggression
  - → Greed nervousness
  - → Excitability

⇒ Example of Unsafe acts

→ Operating the machine of unsafe speed

→ making safety devices inoperative

→ Not using safety devices
→ Horseplay (Teasing, distracting)
→ Starting machine without warning
→ Using Unsafe equipment
→ Crossing a fence
→ maintenance work on moving machines

⇒ Examples of unsafe Condition
→ PPE not available / not in operation
→ moving parts of equipments without guard
→ faulty design of machine
→ use of defective & damaged equipment
→ Unsafe process or method of work
→ poor house keeping.
→ Unsafe working Condition
  - Low illumination
  - Steep roadways
  - Leak in Underground mines.

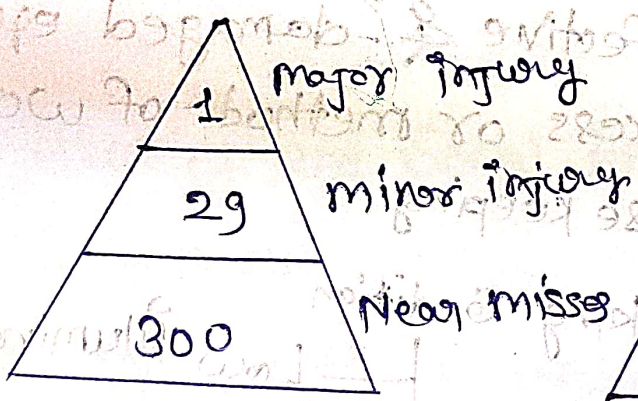
Example:-

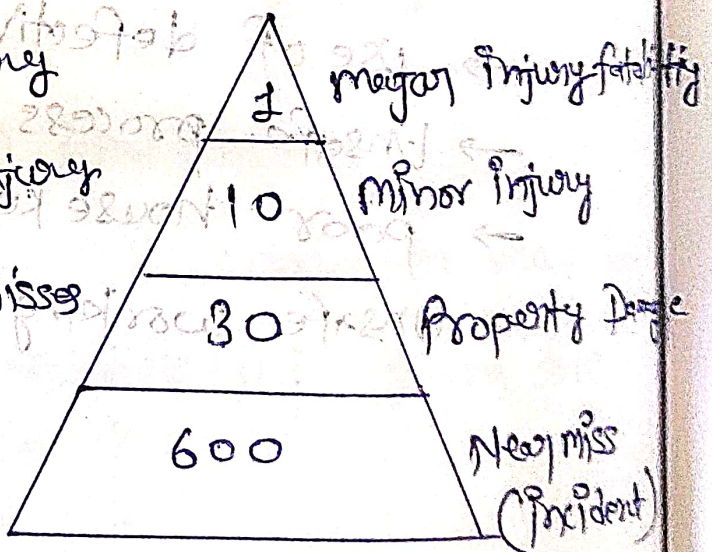
A workey issued a cap-lamp and went into a U/g mine After 3 hrs, the light of cap-lamp Started diminishing. He informed to his Supervisor and the Supervisor suggested to get a new one from Surface. While returning, his

Cap lamp/light completely exhausted and then also he continued to walk in the darkness and ~~Stucked~~ Struck with wooden sleepers and fall down and get injured.

↳ Unsafe act – Returning to surface without any help even after the battery was dead.

⇒ Accident Traingle



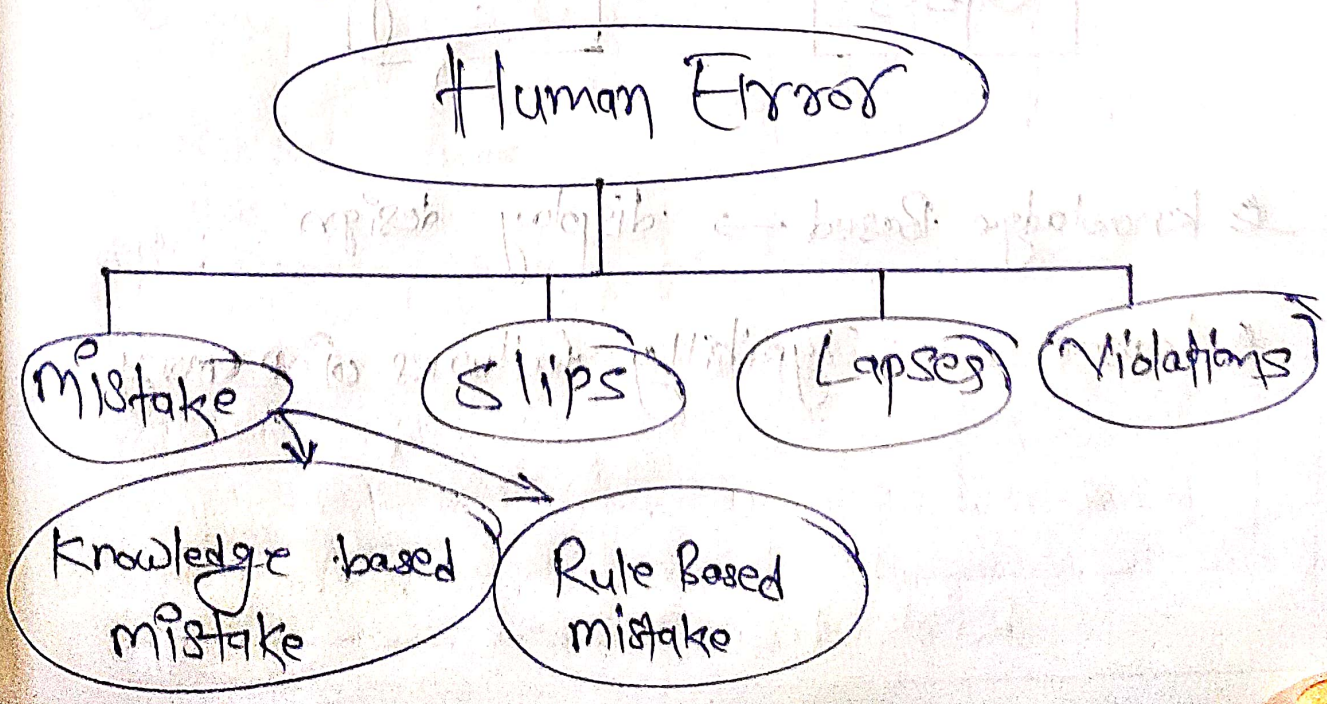


☆ Identification of Humen Errors in Mine

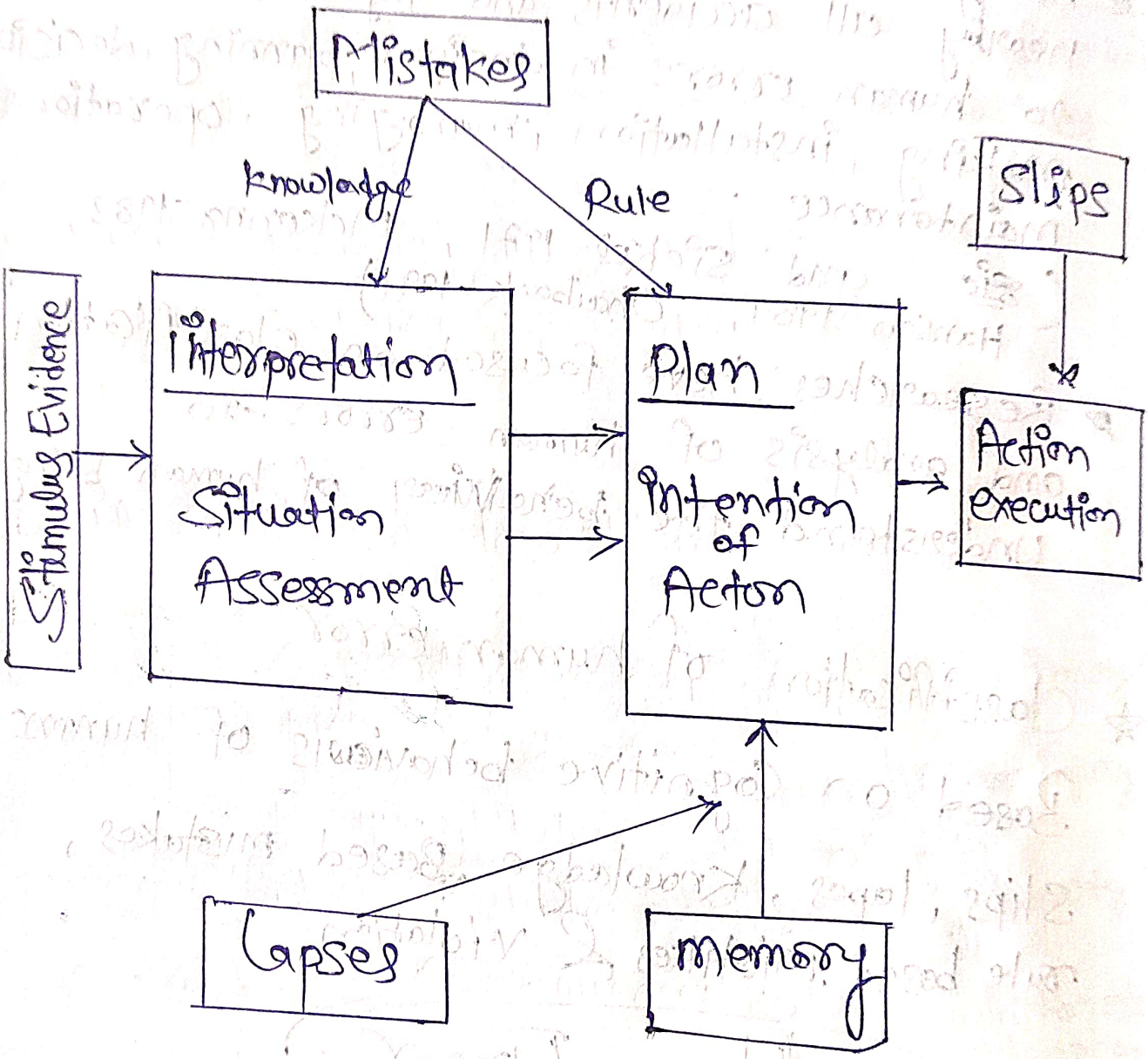
- Analysis of Case studies. indicates that nearly all accidents and injuries occurs due to human errors in design, planning, decision making, installation, managing, operation or maintanance.
  (Sir and Sicked 1971, Mckenna 1983, Hanson 1989, Ohaiback 1991)
- Researches here focused on classification and analysis of human errors to understand the behaviour of human beings

☆ Classification of human error

Based on Cognitive behaviours of humans:
Slips, lapes, Knowledge Based mistakes, rule based mistakes & violation



Taxonomy of Human Error



- Knowledge Based → display design
- Lapses → Typically, failures of memory

Risk Management in Mines

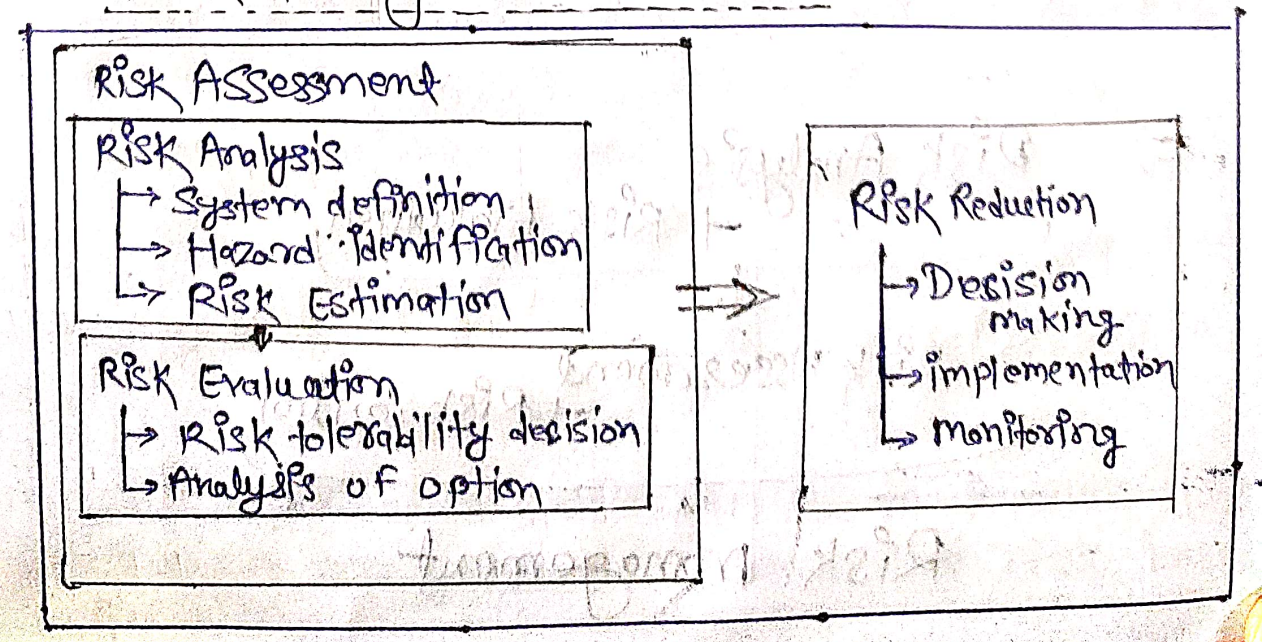
Introduction —

Risk management may be defined as the system & procedures for reducing or eliminating risk including human, organizational, environmental and technological Aspect.

⇒ Risk Management has wide Spread application

- Financial Risk
- Credit Risk
- Project Risk
- Environmental risk
- Safety Risk

Risk Management Process



⇒ Risk Estimation — find losses

- Risk Evaluation — How much loses.

✯ Risk Analysis :—

→ Risk Analysis is a technique for identifying, characterizing, quantifying & evaluating hazards.

→ it consists of two distinct phases —

- A quantitative step of identifying, characterizing & Ranking hazard.
- A quantitative step of risk evaluation which includes estimating the likelihood and consequences of hazard occurance.

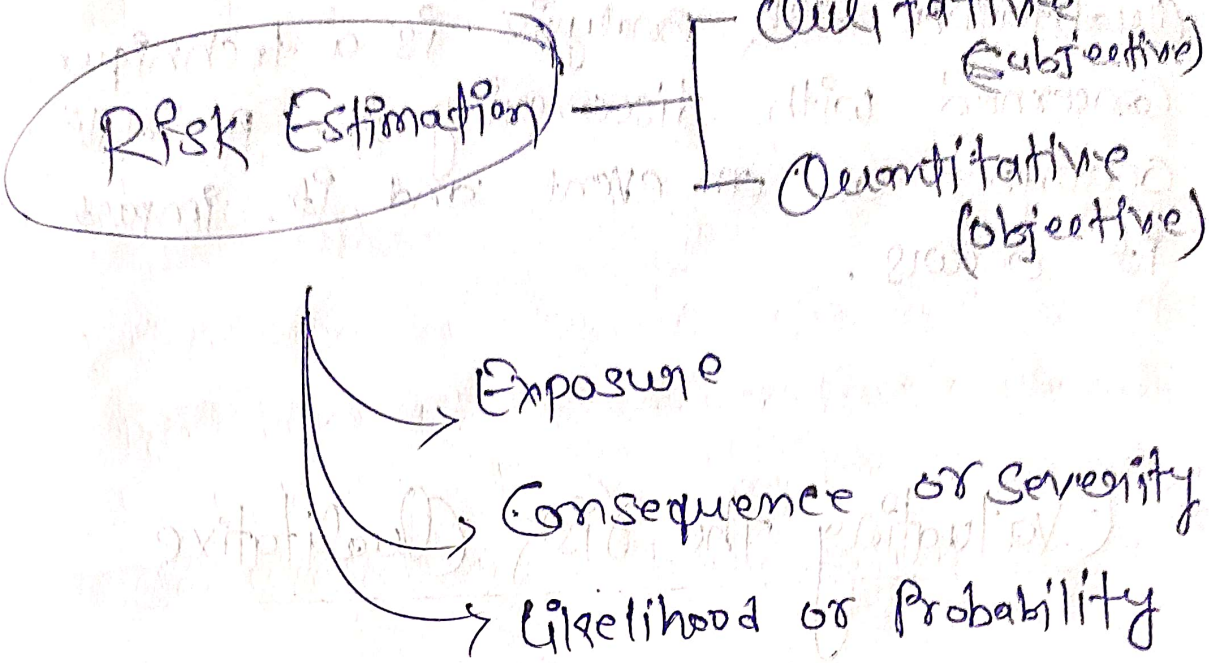
⇒ Risk Analysis + Risk Evaluation

↓

Risk Assessment + Risk Control

↓

Risk management



① Quantitative Risk Analysis —

- Quantitative risk Analysis (QRA) is the art and science of developing and understanding numerical estimates of the risk (i.e. Combination of the expected frequency / prob. / likelihood, and the consequence) associated with a facility or operation.
- It uses a set of highly sophisticated but approximate tools for acquiring and understanding the risk.
- The QRA method can be used throughout all phases of the life of a process or a system.
- The primary goal of QRA is to provide just enough analysis to answer the risk based question.

② Qualitative Risk Analysis —

Qualitative Risk Analysis is a technique concerned with discovering the probability of occurance of an event and its impact if it occurs.

⇒ Evaluating the Risk Qualitative

Likelihood

① Highly unlikely
② Possibility
③ Quite likely
④ Very likely

Severity

① Slight harm
② Injury affecting work
③ Serious injury
④ Possible fatality

Risk Matrix →

Likelihood \ Severity				
	4	8	12	16
	3	6	9	12
	2	4	6	8
	1	2	3	4

Risk = likelihood X Severity

Controlling the Risk —

① Unacceptable — Stop doing it until improvement are made.

② Significant — Proceed with the caution but improvement high priority

③ Tolerable — Okay to proceed, but plan to improve.

④ Insignificant — Any possible improvement low priority

→ Risk Metrix —

Severity

Likelihood				
	4 Tolerable	8 significant	12 Unacceptable	16 Unacceptable
	3 insignificant	6 Tolerable	9 Significant	12 Unacceptable
	2 insignificant	4 Tolerable	6 Tolerable	8 Significant
	1 insignificant	2 insignificant	3 insignificant	4 Tolerable

⇒ Risk Assesment Stages:—

✡ Risk Control

Risk Assesment in combination with risk Control or risk elimination refers to the process of risk management, After assesing the risk suitable control measures are to be ~~evated~~ evaluved.

→ Elimination
→ Substitution
→ Separation
→ Administration
→ training
→ PPE

⇒ Risk Analysis ⇒

✡ Risk Estimation
- Probability
- Consequences

$$Risk = \left[\begin{array}{l}\text{Probability of} \\ \text{Occurance of event}\end{array}\right] \times \left[\text{Consequence}\right]$$

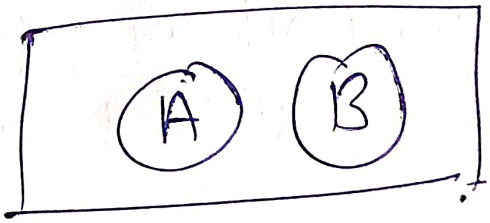
Note:- Success → optimistic (+ve)
failure → pessemistic (-ve)

$$\text{Prob.} = \frac{\text{No. of favourable Outcomes/Event}}{\text{No. of total outcomes/events.}}$$

$$= \frac{f_i}{\sum_{i=1}^{n} f_i}$$

⇒ Mutually Exclusive -

$$P(A \cup B) = P(A) + P(B)$$



⇒ Non-mutually Exclusive -

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

